MD. Saad Bin Sayed

Sixteen Holdings, Apt#6D, 322 Moynarbagh, North Badda, Dhaka, Bangladesh
saad125862@gmail.com +8801998048911 Md Saad Bin Sayed saad125862

Professional Summary

Full-Stack Developer with over two years of professional experience at DeshIT-BD, delivering 24 projects while progressing from AI-Integrated Full-Stack Developer Intern to Mid-Level Full-Stack Developer. Built web and mobile products with NestJS, Next.js, Flutter, PostgreSQL and MySQL, including multi-role ERP, CRM and HR platforms with role-based access control, approval workflows and audit logging, and used Claude Code as a core part of the development workflow. B.Sc. in Computer Science and Engineering from North South University (Fall 2025) and co-author of four international conference papers in AI, machine learning and engineering.

Education

North South University **Fall 2025**
B.Sc in Computer Science and Engineering CGPA: 2.55/4.0

Birshreshtha Munshi Abdur Rouf Public College **2019**
Higher Secondary Certificate (HSC) GPA: 4.08/5.0

Government Laboratory High School **2017**
Secondary School Certificate (SSC) GPA: 5.0/5.0

Experience

DeshIT-BD, Remote (Dhaka, Bangladesh) **Oct 2024 – Oct 2026**

Mid-Level Full-Stack Developer Jul 2026 – Oct 2026

- Owned features from design to release across NestJS, Express, Next.js and Flutter products; reviewed code written by junior developers and shared knowledge within the team.
- Built ten end-to-end projects, including StitchFlow ERP, Distributor ERP, HR Ascend, Dokan ERP and a Sales CRM, using Claude Code as a core part of the workflow to speed up delivery and API optimization.

Junior Full-Stack Developer Feb 2025 – Jul 2026

- Built eleven end-to-end projects with Node.js APIs, user interfaces and MySQL/PostgreSQL databases; wrote tested, documented code and took part in code reviews.

AI-Integrated Full-Stack Developer Intern Oct 2024 – Jan 2025

- Delivered three end-to-end projects in just over three months, each with a NestJS backend and web application, a Flutter Android app, PostgreSQL/MySQL databases and QA testing.
- Used Claude AI to speed up development, optimize backend APIs and prototype interfaces.

Projects

- **CommercePilot AI (Facebook SaaS)** | Next.js, Express, MySQL/TiDB, LLM APIs Oct 2026
  - SaaS with a Messenger AI sales agent and Facebook comment auto-replies: LLM function calling (catalog, stock, shipping, order draft, human handoff) takes customers from comment to confirmed order.
  - Deterministic safety rules and server-side validation; HMAC-verified Meta webhooks, encrypted Page tokens, idempotent replays, per-conversation locks and transactional orders tested under concurrency.
  - Super-admin portal: free trials, plans, invoices, AI usage limits, audit logs; 11 test suites.
- **StitchFlow ERP** | NestJS, Next.js, Flutter, PostgreSQL, Prisma, Docker Sep 2026
  - Garments-manufacturing ERP covering orders, BOM, purchasing, inventory, cutting, sewing, QC, finishing, delivery, finance and maintenance, built on top of the HR Ascend platform.
  - Multi-factory, 18 permission-based roles, two-level approvals, audit log, English/Bangla UI.
  - 170+ data models, 40+ API controllers, 100+ admin pages, Flutter app; Jest and smoke tests.
- **Distributor ERP** | NestJS, Next.js, Flutter, PostgreSQL, Prisma Sep 2026
  - Multi-tenant ERP for distribution companies covering sales, purchase, warehouse and delivery, fleet and logistics, HR and payroll, subscription billing, a customer/vendor portal, CRM and an executive dashboard.
  - Specified a 673-feature, 22-role system from the analysis of a 694-screen prototype; built Phases 0–10 of a 13-phase roadmap across the API, web and mobile apps (50+ data models, 70 web pages).
- **Sales CRM (Lead Desk)** | Flutter, Riverpod, Next.js, PostgreSQL, Prisma Aug – Sep 2026
  - Sales-performance CRM for Sales Executive and Sales Person roles: configurable lead pipeline, lead assignment, deals, tasks, follow-ups, targets, awards, calendar and team analytics.
  - Phone/WhatsApp call flow: post-call summary, follow-ups, recording playback and timeline.
  - Clean-architecture Flutter app (38+ screens) and Next.js API (46 routes, 68 use cases, 15 data models) with JWT auth, OTP password reset by email and Render deployment.

- **HR Ascend** | NestJS, Next.js, Flutter, PostgreSQL, Prisma, Docker Jul – Sep 2026
  - HR and attendance platform for Employee, Manager and HR Admin roles: attendance and leave approvals, expense claims, device approval, digital ID, help desk, announcements and a knowledge base.
  - Clean-architecture NestJS API with JWT access/refresh authentication, Swagger documentation, rate limiting, S3-compatible file storage and PDF export; Next.js admin panel (80 pages) and Flutter mobile app.
- **Dokan ERP** | Express, Next.js, Flutter, PostgreSQL, Prisma, Socket.IO Jul – Aug 2026
  - Shop-management ERP with barcoded products and batch pricing, sales and payments, stock movements, Excel product import with validation reports and subscription-based access.
  - Express/Prisma API with real-time Socket.IO updates, a Next.js web app and an offline-first Flutter app that queues changes and syncs when the API is reachable (about 50 data models).

Research & Publications

- **Harnessing Technology for Climate Resilience: A New Paradigm in Urban Cooling through Eco-friendly Cooler**
  5th International Conference on Environmental Science and Applications (ICESA 2024) (Paper No. 132). Lisbon, Portugal: ICESA.
  Authors: Ahmed Faizul Haque Dhrubo, Md. Saad Bin Sayed, Labiba Faizah, Samiur Rahman Alif, Md. Mohibul Islam, Farhana Sharmin, Mohammad Abdul Qayum
- **Leveraging Advanced AI for Real-Time Sign Language Detection and Translation: A Pioneering Approach to Bridging Communicative Gaps and Promoting Societal Inclusivity**
  1st IEEE Conference on Secure and Trustworthy CyberInfrastructure for IoT and Microelectronics (SaTC 2025)
  Authors: Sk. Md. Towfikul Islam Siam, Maher Ali Rusho, Md. Saad Bin Sayed, Md. Mofazzal Hossen, Sumiya Sultana, Mehedi Hasan Refat, Ahmed Faizul Haque Dhrubo, Mohammad Abdul Qayum
- **Defending Against Evasion Attacks in Adversarial Machine Learning: An Optimization Analysis and Framework**
  4th International Conference on Computing and Machine Intelligence (ICMI 2025)
  Authors: Tashfin Rahat Ratul, Samiur Rahman Alif, Maher Ali Rusho, Sk. Md. Towfikul Islam Siam, Hadi Uzzaman, Md. Saad Bin Sayed, Ahmed Faizul Haque Dhrubo, Md. Khurshid Jahan
- **Advancing Sentiment Analysis with BERT: A Comparative Evaluation Against Machine Learning and Deep Learning Techniques**
  2026 IEEE 2nd International Conference on Secure IoT, Assured and Trusted Computing (SATC), March 2026
  Authors: Md. Saad Bin Sayed, Maher Ali Rusho, Zarin Tasnia Shanta, Raiyan Rafsan, Tashfin Rahat, Ashfaqur Rahman Chowdhury, Ahmed Faizul Haque Dhrubo, Mohammad Abdul Qayum

Technologies

Languages: TypeScript, JavaScript, Dart, Python, Java, C, C++, PHP, Assembly, HTML

Backend, Web & Mobile: NestJS, Express, Next.js, React, Flutter, Prisma ORM, REST APIs, Swagger/OpenAPI, JWT Authentication, Socket.IO, Zod

Databases: PostgreSQL, MySQL (TiDB Cloud), MongoDB, Oracle SQL, Microsoft SQL

DevOps, Testing & Tools: Docker, Docker Compose, Render, Vercel, Neon, Backblaze B2 (S3), cPanel (Passenger), Jest, smoke testing, QA testing, Git, GitHub, VS Code, Jupyter Notebook, LaTeX

AI-Assisted Development: Claude Code, agent-driven workflows, project instruction files (CLAUDE.md)

Machine Learning & Data Science: Scikit-learn, Pandas, NumPy, TensorFlow, K-Means Clustering, t-SNE

Generative AI & LLM Engineering: OpenAI API, DeepSeek/Gemini APIs, LLM Function Calling, CrewAI, n8n, Prompt Engineering, AI Agents, RAG, Vector Databases, LLM Fine-tuning

IoT & Microcontrollers: Arduino, STM32, ESP WiFi Module, HC-06 Bluetooth, DHT22 Sensor

Certificates

Generative AI Engineering Mastermind
Outskill 2026
Completed intensive 2-day hands-on workshop focused on building 30 production grade AI applications, multi-agent systems, and LLM engineering.

Extracurricular Activities

North South University Computer and Engineering Club (NSU CEC) Member, Fall 2024

References

Professor Dr. Mohammad Abdul Qayum
Assistant Professor
Dept. of Electrical and Computer Engineering
North South University, Dhaka-1229, Bangladesh
mohammad.qayum@northsouth.edu

Professor Dr. Md. Sazzad Hossain
Adjunct Professor
Dept. of Electrical and Computer Engineering
North South University, Dhaka-1229, Bangladesh
sazzad.hossain14@northsouth.edu